45A, 600V Low Loss Single-Phase Single In-Line Bridge Rectifier

FEATURES

- Qualification reference AEC-Q101
- Low forward drop enhances the efficiency
- High surge current capability
- UL Recognized File # E-326243
- High case dielectric strength of 2500 V_{RMS}
- RoHS Compliant
- Halogen-free

APPLICATIONS

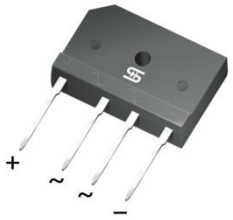
- Switching mode power supply (SMPS)
- AC to DC
- On board charger

MECHANICAL DATA

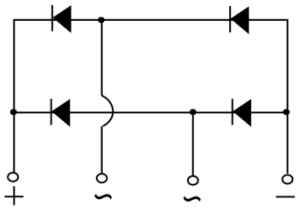
- Case: TS-6P
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: As marked
- Weight: 7.27g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	45	A
V_{RRM}	600	V
I_{FSM}	620	A
$T_{J\ MAX}$	175	°C
Package	TS-6P	
Circuit Configuration	In-line	

Pb-Free RoHS COMPLIANT HALOGEN FREE



TS-6P



ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**VALUE**	**UNIT**
Repetitive peak reverse voltage		V_{RRM}	600	V
Reverse voltage, total rms value		$V_{R(RMS)}$	420	V
Forward current		I_F	45	A
Surge peak forward current, single half sine-wave superimposed on rated load per diode	t = 8.3ms	I_{FSM}	620	A
	t = 1.0ms		1200	A
Rating of fusing (t<8.3ms)		I^2t	1595	A^2s
Junction temperature		T_J	- 55 to +175	°C
Storage temperature		T_{STG}	- 55 to +175	°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-lead thermal resistance	$R_{\Theta JL}$	0.8	°C/W
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	4.6	°C/W
Junction-to-case thermal resistance	$R_{\Theta JC}$	0.6	°C/W

Thermal Performance Note: Mounted on heat sink with 4" x 6" x 0.25" Al-Plate

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage per diode(1)	I_F = 22.5A, T_J = 25°C	V_F	0.91	0.95	V
	I_F = 22.5A, T_J = 125°C		0.79	-	V
Reverse current @ rated V_R per diode(2)	T_J = 25°C	I_R	-	10	μA
	T_J = 125°C		0.9	-	μA
Junction capacitance per diode	1MHz, V_R = 4.0V	C_J	382	-	pF
Reverse recovery time	I_F = 0.5A, I_R = 1.0A, I_{rr} = 0.25A	t_{rr}	1243	-	ns

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING
GBJL45J	TS-6P	15 / Tube

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

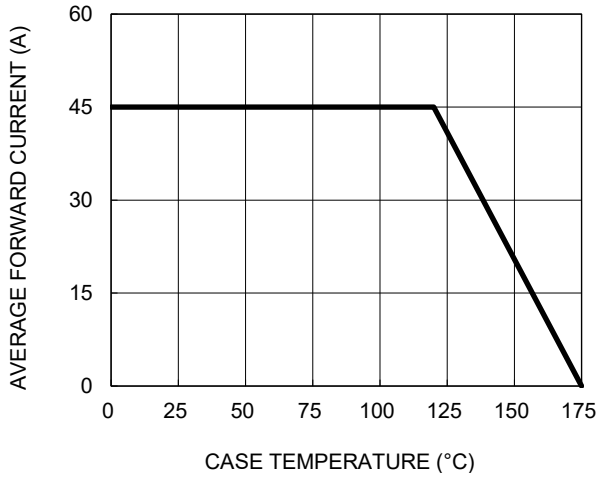


Fig.2 Typical Junction Capacitance

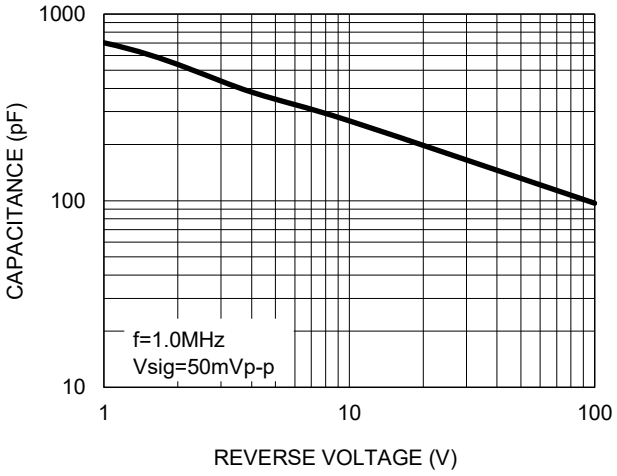


Fig.3 Typical Reverse Characteristics

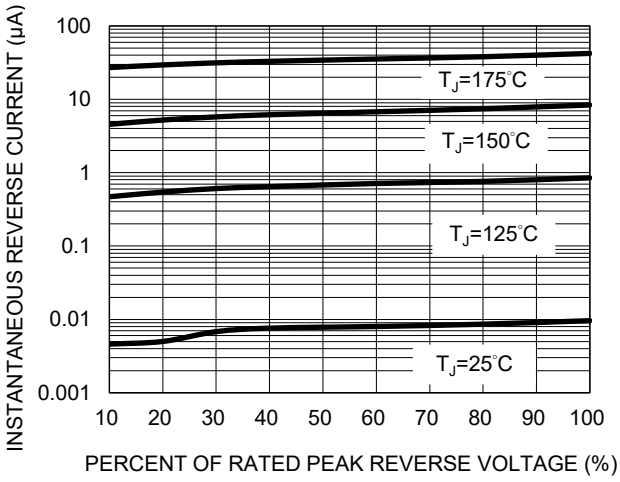


Fig.4 Typical Forward Characteristics

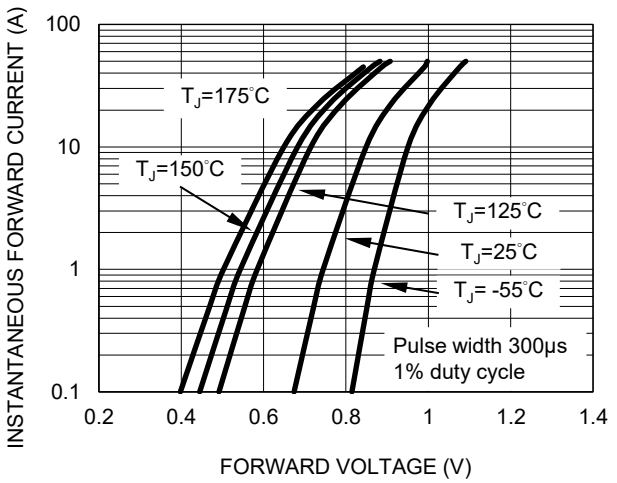
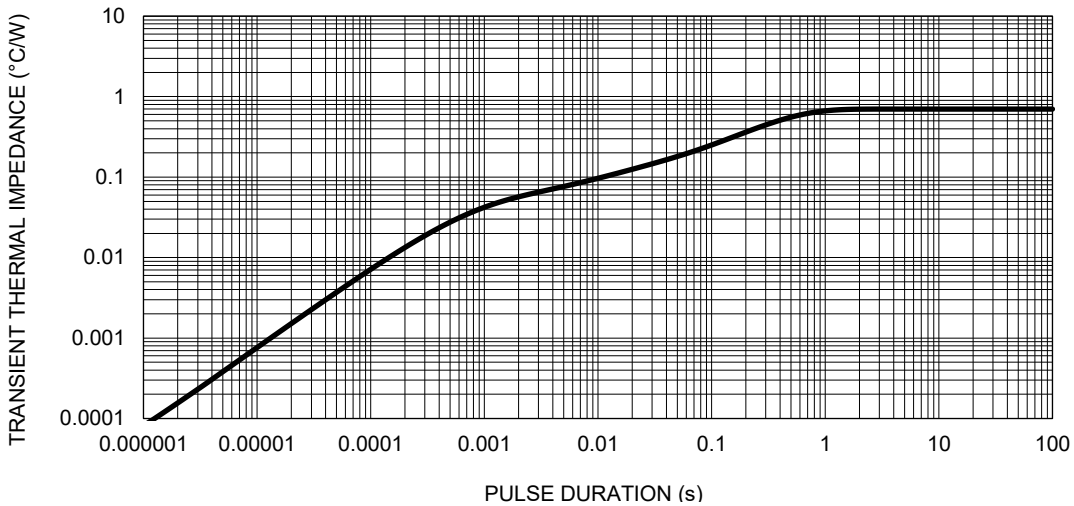
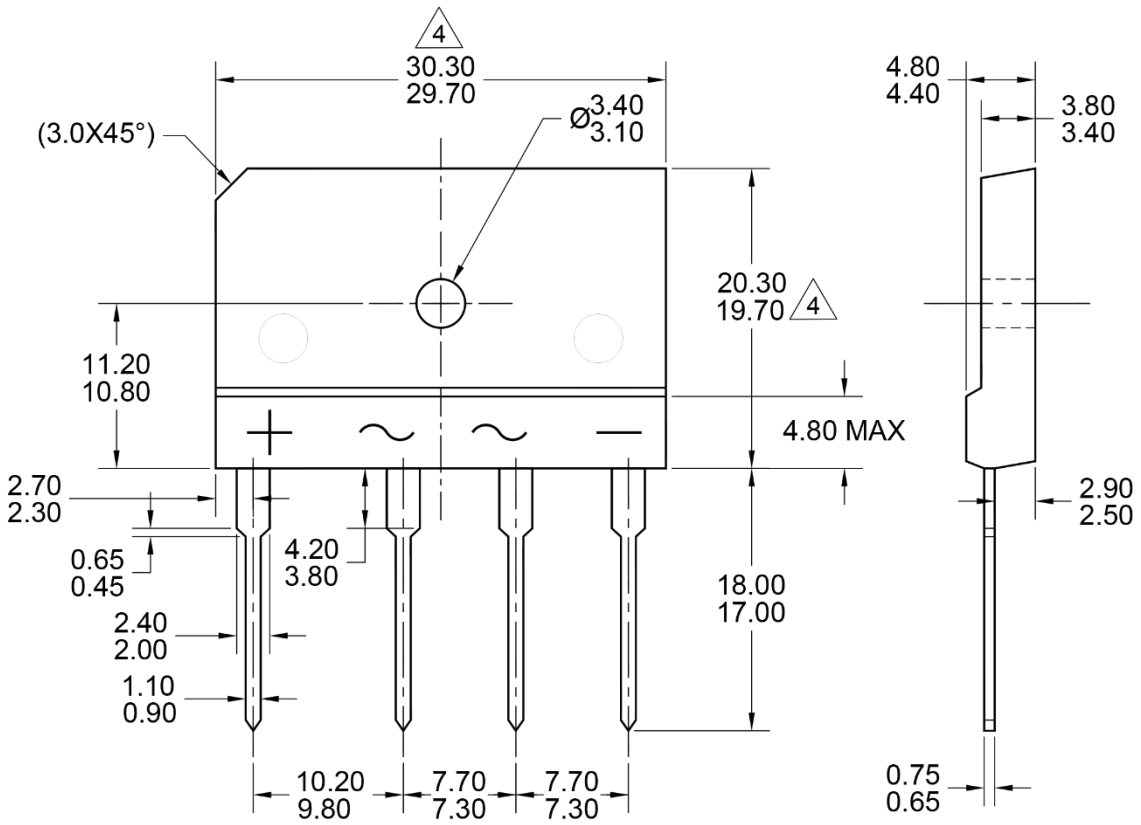


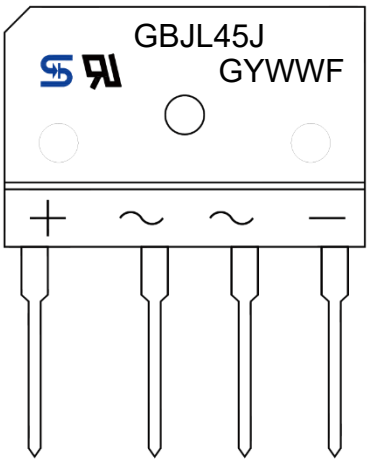
Fig.5 Typical Transient Thermal Impedance



PACKAGE OUTLINE DIMENSIONS

TS-6P





MARKING DIAGRAM

G = GREEN COMPOUND
YWW = DATE CODE
F = FACTORY CODE

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. THERE IS NO EXISTING PACKAGE OUTLINE INDUSTRY STANDARD FOR THIS PACKAGE.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-TS6P-069 REV A.

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